Software Requirements Specification (SRS)

Medical Appointment Web System

Subject: Advanced Web Development

NRC:

Team 6

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1. Introduction

1.1. Purpose

The purpose of this document is to define the functional and non-functional requirements for the *Medical Appointment Web System*. The goal is to design a centralized, web-based platform that allows patients, doctors, and administrators to efficiently manage medical appointments, improving patient satisfaction and clinic productivity.

1.2. Scope

The system provides an integrated platform that enables:

- Patients: to book, manage, and track appointments, receive reminders, and access medical history.
- Doctors: to manage schedules, access patient health records, and document consultation notes.
- Administrators: to handle user and resource management, monitor performance, and generate reports.

The system will be developed as a web application using React (frontend) and PostgreSQL (database), with secure APIs for data communication.

1.3. Definitions, Acronyms and Abbreviations

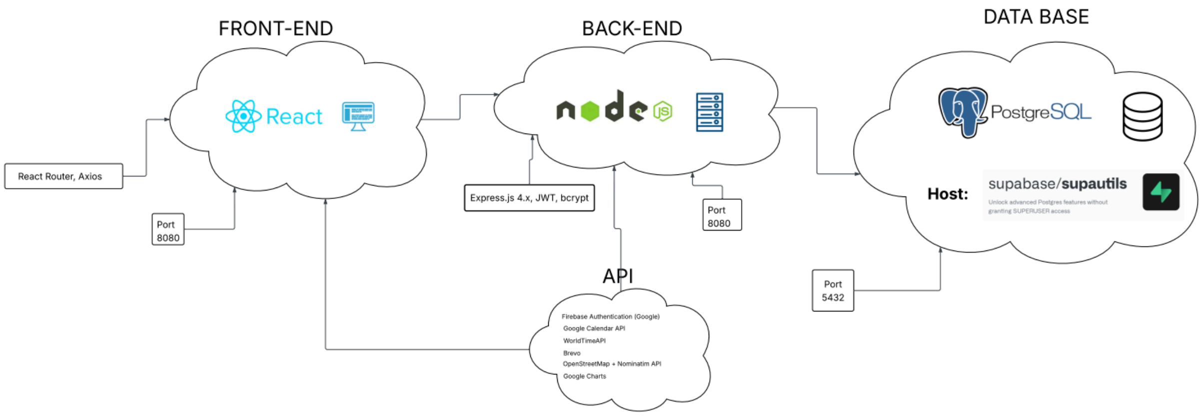
Term	**Definition**
EHR	Electronic Health Record
CRUD	Create, Read, Update, Delete
RBAC	Role-Based Access Control
API	Application Programming Interface
SSO	Single Sign-On

2. Overall Description

2.1. Product Perspective

The system follows a three-tier architecture:

- Frontend (React 18+) – provides user interface and client logic.
- Backend (API) – handles business logic and data management.
- Database (PostgreSQL) – stores user, appointment, and medical data.



2.2. Product Functions

Main functionalities per user type:

User	Functionality
Patients	Register, authenticate, search for available appointments, book, modify, or cancel appointments, receive reminders, submit satisfaction surveys, and access EHR.
Doctors	Set availability, view consolidated calendars, access EHRs, update appointment statuses, and add consultation notes.
Administrators	Manage users and resources, monitor system metrics, generate reports and invoices, and access audit logs.

2.3. User Characteristics

User	Functionality
Patients	Basic computer and web navigation skills
Doctors	Intermediate users familiar with medical systems
Administrators	Technical or administrative personnel with full system access.

2.4. Constraints

- Must support 100+ concurrent users.
- Search operations ≤ 3 seconds.
- 99.5% uptime availability.
- Must comply with HTTPS/TLS standards.

2.5. Assumptions and Dependencies

- Internet connection is required.
- External APIs: Firebase (Auth), Brevo (Notifications), Google Calendar (Sync).
  All users must verify accounts through email or Google SSO.

3. Specific Requirements

3.1. Functional Requirements

ID	Description	Priority
RF-P.1	Patient registration with email verification	High
RF-P.2	Authentication via credentials or Google SSO	High
RF-P.3	Search appointments by doctor, specialty, or date	High

RF-P.4	Automatic time slot calculation	High
RF-P.5	Appointment booking and confirmation	High
RF-P.6	Modify or cancel appointments	High
RF-P.7	Automatic reminders (24h and 1h before)	Medium
RF-P.8	Post-appointment satisfaction survey	Medium
RF-D.1	Configure doctor schedules and breaks	High
RF-D.2	View consolidated calendar	High
RF-D.3	Access patient EHR	High
RF-D.4	Update appointment status	High
RF-D.5	Add diagnostic notes	High
RF-A.1	User CRUD and role assignment	High
RF-A.2	Manage consultation rooms and equipment	High
RF-A.3	Generate performance reports	Medium
RF-A.4	Create billing invoices	Medium
RF-A.5	Access audit logs	Medium

4. System Models

Non-Functional Requirements

ID	Category	Description	Target
RNF-P.1	Performance	Appointment search $\leq$ 3s	Met
RNF-P.2	Scalability	Support for 100+ users	Met

RNF-P.3	Availability	99.5% uptime	Met
RNF-S.1	Security	JWT-based authentication	Met
RNF-S.2	Security	HTTPS/TLS and data encryption	Met
RNF-S.3	Security	Role-based access control (RBAC)	Met
RNF-C.1	Usability	Intuitive and responsive UI	Met
RNF-C.2	Maintainability	Clean, well-documented code	Met
RNF-C.3	Portability	Support major browsers	Met

4.1. Main Entities

- User: id, name, email, password, role
- Patient: date_of_birth, insurance_plan
- Doctor: specialty, license_number
- Appointment: datetime, status, duration
- MedicalRecord: allergies, diagnoses, prescriptions
- ConsultationRoom, DoctorSchedule, Reminder, Billing, AuditLog, Survey

4.2. Appointment Lifecycle

Scheduled → Confirmed → In Progress → Completed/Cancelled

5. External Interfaces

Interface	Description	Provider
Authentication	Login via Google SSO	Firebase

Calendar	Doctor schedule synchronization	Google Calendar API
Notifications	Appointment reminders (email/SMS)	Brevo (Sendinblue)
Reports	Charts and visual analytics	Google Charts

6. Appendices

6.1. Future Enhancements

- Payment gateway integration (Stripe, PayPal).
- Mobile application version.
- Predictive analytics for appointment optimization.

7. Analyst Comparison

To evaluate the performance and efficiency of the medical appointment system, as well as the professionals who use it, various indicators can be compared among doctors, specialties, or administrative staff. Each doctor operates within the same digital environment, but their performance may vary depending on the number of patients attended, appointment punctuality, and patient satisfaction.

The system records detailed data from every appointment, including scheduled time, check-in time, consultation duration, cancellations, and follow-ups. Using these records, several performance metrics can be computed:

- Total Consultations (TC): total number of appointments completed by a doctor or specialty.
- Attendance Rate (AR): ratio of completed appointments to scheduled appointments.
- Average Consultation Time (ACT): average duration of a consultation.
- Cancellation Ratio (CR): percentage of appointments cancelled or rescheduled by patients or staff.

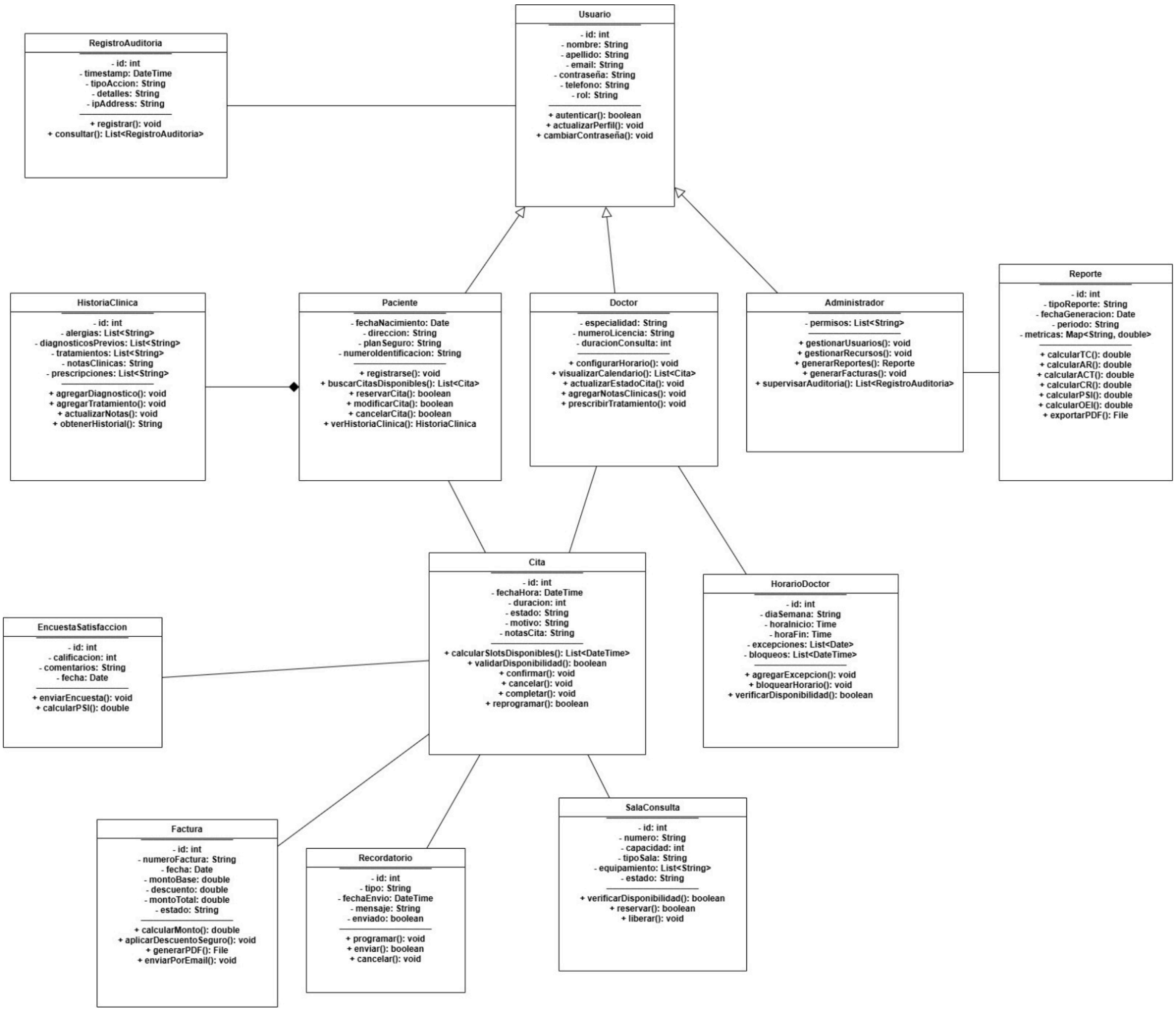
- Patient Satisfaction Index (PSI): rating derived from patient feedback surveys after each visit.

By analyzing these metrics, the clinic can identify patterns such as which doctors maintain the highest patient satisfaction, which specialties experience more cancellations, or which time slots cause delays. This data-driven comparison supports administrative decisions like workload balancing, incentive programs, and scheduling adjustments.

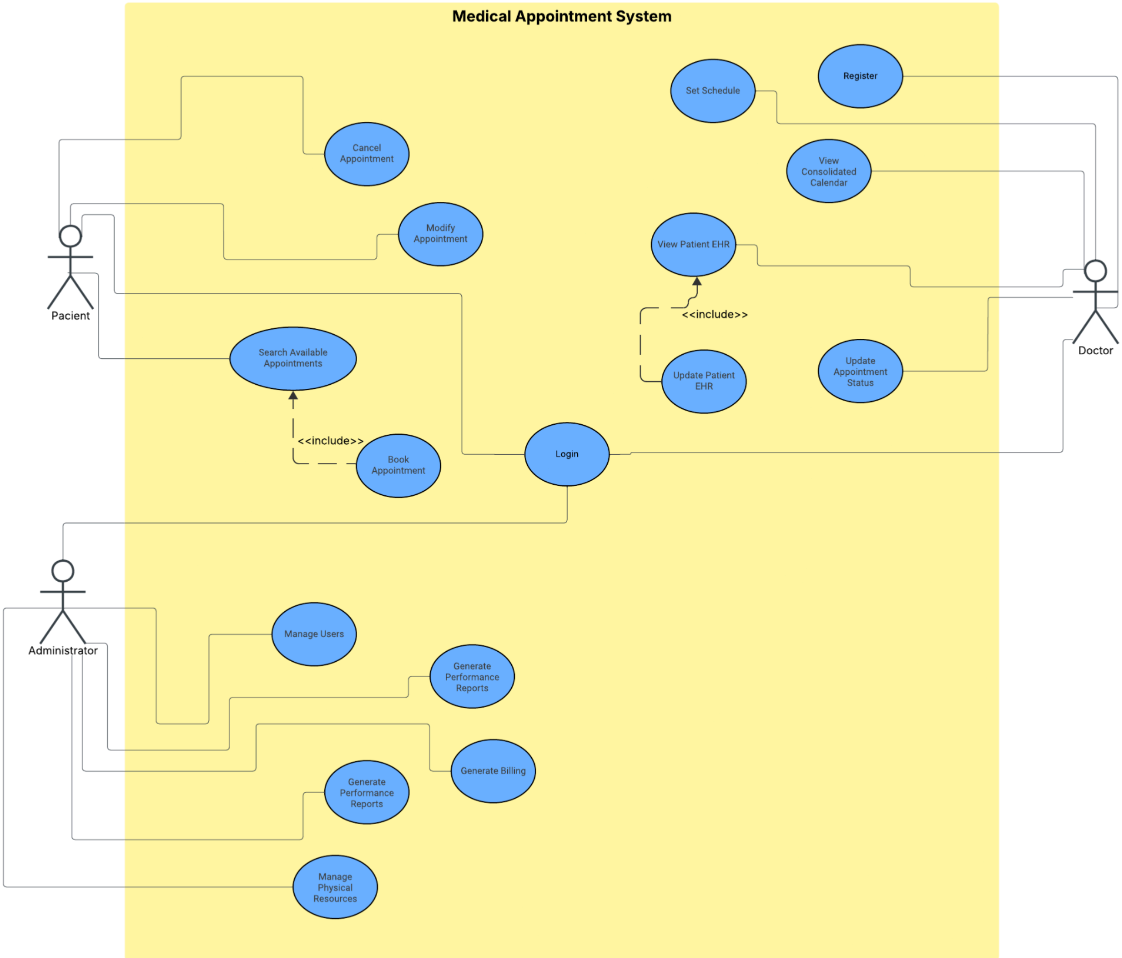
Another measure of overall clinic performance is the Operational Efficiency Index (OEI), calculated as the ratio between the total number of effective consultations and the total available time of medical staff. A higher OEI indicates that the system and its users are utilizing resources optimally.

In summary, this comparison framework allows the clinic to continuously improve its service quality, identify inefficiencies, and enhance the patient experience through evidence-based management.

8. Class Diagram



9. Use Case Diagram



10. E-R Model

https://supabase.com/dashboard/project/nfzvqtfhoqdxvzigtxje/database/schemas